Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim: Open ended problem-7**	
Open ended - 7	**Date:12/08/2026**	**Enrollment No:92400133057**

Github:-https://github.com/manavkasundra/DSIP

Primary Goal:-

To improve the visual quality of five distinctly degraded grayscale images by applying spatial domain point-processing operations.

Source Images:-

- A picture of seeds that looked completely faded and washed out.
- A circular texture that was so visually flat, its pixel values barely varied.
- Another batch of seeds that was so underexposed, everything was lost in deep shadows.
- A nighttime photo of a car, which is always tough because you have to balance harsh lights with pitch blackness.
- An abdominal CT scan that was too dark to read due to improper medical display settings.

Techniques Utilized:-

To enhance these photos, we applied three classic image-editing math tricks:

- **Linear Contrast Stretching:** Pulling the darks and lights further apart.
- **Gamma Correction:** Tweaking the mid-tones to naturally brighten or darken the scene.
- **Histogram Equalization:** Redistributing the pixels so the image has a perfectly balanced range of darks and lights.

Results & Drawbacks:-

- Successes: Each processed image demonstrated substantial visual upgrades, successfully uncovering previously obscured textures, structural outlines, and internal anatomical details.
- Main Limitation: Because these are global point-processing operations, they tend to exaggerate existing sensor noise and struggle to retain bright highlight details within high-dynamic-range scenes.

Defining a Digital Image:

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- A digital image functions as a two-dimensional grid of discrete values, mathematically represented as *f(x, y)*.
- *x* represents the horizontal position (the column).
- *y* represents the vertical position (the row).
- *f* represents the specific gray-level intensity at that exact *(x, y)* coordinate.

Grayscale Image Parameters:

- Standard grayscale formats operate using an 8-bit color depth.
- Minimum Limit (0): Indicates pure black (a total absence of light).
- Maximum Limit (255): Indicates pure white (maximum light intensity).
- Total Range: Encompasses 256 distinct shades of gray from pure black to pure white.

Why Do Images Degrade?

Images can lose their quality or become unclear for several reasons:

- **Environmental factors:** Poor lighting, shadows, or differences between daytime and nighttime can affect image quality.
- **Hardware limitations:** Cameras have limitations such as sensor noise and a limited ability to capture very bright and very dark areas at the same time.
- **Operator mistakes:** Incorrect exposure settings can make an image too bright or too dark.
- **Medical imaging errors:** In radiology, incorrect **window and level settings** can make important details in a scan difficult to see.

Introduction to the Image Histogram

An **image histogram** is a graph that shows how the brightness levels are distributed throughout an image.

- **X-axis:** Shows the different gray levels, from **0 to 255**.
  - 0 represents **black**.
  - 255 represents **white**.
- **Y-axis:** Shows how many pixels in the image have each particular gray level.

How to Understand a Histogram

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- **Narrow peak in the middle:** Usually indicates a **low-contrast image**, which may look dull or washed out.
- **Large peak near 0:** Means that most pixels are very dark, so the image is likely **underexposed**.
- **Peaks near both 0 and 255:** Indicates that the image contains a large range of dark and bright pixels, which can happen in scenes with **very high contrast**, such as nighttime images with bright lights.

Introduction to Point Processing (Gray-Level Operations)

Point processing is a method of changing an image where the new value of each pixel depends only on the original value of that same pixel.

It can be represented as:

s = T(r)

Here, **r** is the original pixel value and **s** is the new pixel value after processing.

Advantages

- It is **simple and fast** because each pixel is processed independently.
- It requires relatively little computation, with approximately **O(N)** complexity.
- It can be easily implemented using both **software and hardware**.

Disadvantages

- It does not look at the pixels around a particular pixel, so it cannot perform operations such as **blurring or sharpening**.
- It has **no spatial awareness**, meaning every pixel is processed individually without considering its surroundings.

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Technique 1: Linear Contrast Stretching

Purpose:

Linear contrast stretching is used to improve the contrast of an image. It takes a small or narrow range of gray values and spreads them across the complete **0–255 range**. This makes dark areas darker, bright areas brighter, and the overall image easier to see.

Mathematical Formula:-

$$s = \left(\frac{r - r_{min}}{r_{max} - r_{min}} \right) \times 255$$

Where:

- **R min:** The darkest pixel value present in the original image.
- **R max:** The brightest pixel value present in the original image.
- **r:** The original gray level of a pixel.
- **s:** The new gray level after stretching.

In other words, the formula takes the existing gray levels and spreads them out between 0 and 255.

Graphical Interpretation:-

The transformation function **T(r)** is represented by a straight diagonal line.

Slope Analysis:-

If the difference between r max and r min is very small, it means the image has a narrow range of gray levels and therefore low contrast.

In this situation, the slope of the transformation line becomes steep (greater than 1). As a result, even small differences in the original pixel values become larger, which significantly improves the image contrast.

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In short: Linear contrast stretching takes a dull, low-contrast image and spreads its intensity values across the full 0–255 range, making the image clearer and more detailed.

Technique 2: Power-Law (Gamma) Transformation

Purpose:-

Power-law or gamma transformation is used to adjust the brightness of an image in a way that is closer to how the human eye perceives brightness. It is especially useful when an image is too dark or too bright.

Mathematical Formula

$$s = c \cdot r^{\gamma}$$

Where:-

- **r:** Original pixel intensity.
- **s:** New pixel intensity after the transformation.
- **c:** A constant used for scaling, usually taken as **1**.
- **γ:** Gamma value that controls how much the brightness is changed.

Effect of Gamma Value

When $\gamma<1$

For example: 0.3, 0.4, or 0.5

- The curve bends upward.
- Dark areas of the image become much brighter.
- Bright areas are compressed so they do not increase as much.
- This is useful for dark or underexposed images, where important details are hidden in the shadows.

Example: A photo that looks too dark can be brightened using a gamma value below 1.

When $\gamma>1$

For example: 2.0 or 3.0

- The curve bends downward.

- Dark areas become darker or are compressed.
- Bright areas are given more range.
- This can be useful for overexposed or overly bright images.

In simple words:
Gamma transformation allows us to control image brightness by changing the gamma value. A gamma below 1 generally makes the image brighter, while a gamma above 1 generally makes it darker.

Technique 3: Histogram Equalization (HE)

Purpose:
Histogram Equalization is used to improve the overall contrast of an image by redistributing its gray-level values over a wider range.

Basic Idea

Sometimes, most of the pixels in an image are concentrated around only a few gray levels. This can make the image look dull and hide important details.

Histogram equalization spreads these commonly occurring gray levels across a larger range. Less frequently occurring gray levels may be compressed.

As a result, the image usually gets better contrast and more visible details.

Application

Histogram equalization is particularly useful when an image has a very narrow or highly concentrated histogram and contains details that are difficult to see.

For example, it can help reveal hidden textures and features in very dark or low-contrast images.

In simple words:
Histogram equalization takes an image where most pixels have similar brightness levels and spreads those brightness levels out, making the hidden details easier to see.

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Advanced Mathematical Derivation of Histogram Equalization

Step 1: Probability Density Function (PDF):-

The Probability Density Function (PDF) tells us how often each gray level appears in an image. In simple terms, it gives the probability of finding a pixel with a particular gray value.

The formula is:

$$p_r(r_k) = \frac{n_k}{M \times N}$$

Where:

- **pr(rk):** Probability of gray level rk.
- **nk:** Number of pixels that have the gray level rk.
- **M×N:** Total number of pixels in the image.

So, the PDF is basically the Number of pixels with a particular gray level ÷ Total number of pixels.

Step 2: Cumulative Distribution Function (CDF)

The Cumulative Distribution Function (CDF) adds up the probabilities of all gray levels from the beginning up to the current gray level.

It tells us the total accumulated probability up to a particular gray level.

The formula is:

$$T(r_k) = \sum_{j=0}^{k} p_r(r_j)$$

Here:

- **T(rk):** CDF value for the gray level rk.

- **pr(rj):** Probability of each gray level from 0 up to k.
- The summation adds all these probabilities together.

The CDF starts with a value close to 0 and gradually increases until it reaches 1 at the highest gray level, which is 255 for an 8-bit image.

In simple words, the CDF keeps a running total of the pixel probabilities as we move from dark to bright gray levels.

Step 3: Gray-Level Remapping

After calculating the CDF, its values are between 0 and 1. However, an 8-bit image uses gray levels from 0 to 255.

Therefore, we convert the CDF values into the 0–255 range using:

$$s_k = \text{round}(255 \times T(r_k))$$

Where:

- **T(rk):** CDF value of the original gray level.
- **255:** Maximum gray level for an 8-bit image.
- **sk:** New gray level after histogram equalization.
- **round():** Rounds the result to the nearest whole number.

Overall Idea

In simple terms, histogram equalization works in three main steps:

1. **PDF:** Find how frequently each gray level occurs.
2. **CDF:** Add these probabilities cumulatively from dark to bright levels.
3. **Remapping:** Use the CDF to assign new gray levels between 0 and 255.

This process spreads the intensity values over a wider range, which generally improves contrast and makes hidden details more visible. The resulting histogram becomes approximately uniform, although it is not necessarily perfectly flat because the output gray levels are discrete integers.

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How to Pick the Right Tool (The Simple Guide)

Not sure which math trick to use on your photo? Just ask yourself what is wrong with the picture, and use this simple cheat sheet:

1. The Foggy, Boring Photo

- **The Problem:** The picture looks washed out and totally gray. There are no true blacks and no bright whites.
- **The Tool to Use:** Linear Contrast Stretching.
- **Why it Works:** Think of it like a rubber band. It grabs all those boring middle-grays and stretches them out to the edges. Suddenly, your darks are actually dark, your lights are light, and the picture pops.

2. The Super Dark Photo

- **The Problem:** The picture is buried in shadows. You can barely see anything because it is just too dark.
- **The Tool to Use:** Gamma Correction (turned down below 1).
- **Why it Works:** It acts like a "smart flashlight." It brightens up the dark corners so you can see the hidden details, but it is smart enough to leave the bright spots alone so they don't turn blindingly white.

3. The Flat Blob

- **The Problem:** The whole image looks like one flat, featureless blob of gray. You know there are textures or shapes there, but you just can't see them.
- **The Tool to Use:** Histogram Equalization.
- **Why it Works:** It takes that one stubborn clump of gray and forces it to break apart and spread evenly into lots of different shades. It is the ultimate trick for magically revealing completely hidden details.

4. The "Bright Lights, Pitch Black" Photo

- **The Problem:** You have extreme darks and extreme brights in the exact same photo (like bright car headlights on a pitch-black street).
- **The Tool to Use:** Gamma Correction first, then a little bit of Contrast Stretching.
- **Why it Works:** It is a team effort. This combo focuses on lighting up the dark background so you can actually see what is hiding there, even if it means the streetlights get a tiny bit blurry or too bright in the process.

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Code:-

```
import cv2

import numpy as np

import os

def ensure_uint8(image):

   """Detects 16-bit PNGs and safely converts them to 8-bit 0-255."""

   if image is None:

       return None

   print(f" -> Input dtype: {image.dtype}, Min: {np.min(image)}, Max: {np.max(image)}")

   if image.dtype == 'uint16':

       print(" ⚠️ Detected 16-bit PNG. Scaling down to 8-bit...")

       image = (image / 256.0).astype('uint8')

   elif image.dtype != 'uint8':

       image = cv2.normalize(image, None, 0, 255, cv2.NORM_MINMAX).astype('uint8')

   print(f" -> Converted to dtype: {image.dtype}, Min: {np.min(image)}, Max: {np.max(image)}")

   return image

# ==========================================

# 2. ENHANCEMENT FUNCTIONS

# ==========================================

def gamma_correction(image, gamma=1.0):

   inv_gamma = 1.0 / gamma
```

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```
    table = np.array([((i / 255.0) ** inv_gamma) * 255 for i in np.arange(0, 256)]).astype("uint8")

    return cv2.LUT(image, table)

def contrast_stretching(image, lower_percent=2, upper_percent=98):

    min_val, max_val = np.percentile(image, (lower_percent, upper_percent))

    if max_val - min_val == 0:

        return image

    stretched = (image - min_val) * (255.0 / (max_val - min_val))

    return np.clip(stretched, 0, 255).astype(np.uint8)

def clahe_enhancement(image, clip_limit=2.0, grid_size=(8, 8)):

    clahe = cv2.createCLAHE(clipLimit=clip_limit, tileGridSize=grid_size)

    return clahe.apply(image)

# Update to your exact .png file names

input_files = {

    'img1': 'image1.png', # Washed-out Seeds

    'img2': 'image2.png', # Circular Pattern

    'img3': 'image3.png', # Dark Seeds

    'img4': 'image4.png', # Night Car

    'img5': 'image5.png'  # CT Scan

}

print("\n--- Starting ROBUST Enhancement Pipeline ---\n")

# --- IMAGE 1: Contrast Stretching ---
```

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```
if os.path.exists(input_files['img1']):

    img1 = cv2.imread(input_files['img1'], cv2.IMREAD_GRAYSCALE)

    img1 = ensure_uint8(img1)

    img1_out = contrast_stretching(img1, lower_percent=2, upper_percent=98)

    cv2.imwrite('image1_enhanced.png', img1_out)

    print(f"✓ Enhanced {input_files['img1']} -> image1_enhanced.png\n")

# --- IMAGE 2: Stretch FIRST, THEN CLAHE to avoid black-out ---

if os.path.exists(input_files['img2']):

    img2 = cv2.imread(input_files['img2'], cv2.IMREAD_GRAYSCALE)

    img2 = ensure_uint8(img2)

    # Stretch the extremely flat image first to give CLAHE data to work with

    img2_stretched = contrast_stretching(img2, lower_percent=1, upper_percent=99)

    # Then apply CLAHE to get the rings clearly

    img2_out = clahe_enhancement(img2_stretched, clip_limit=2.0, grid_size=(8, 8))

    cv2.imwrite('image2_enhanced.png', img2_out)

    print(f"✓ Enhanced {input_files['img2']} -> image2_enhanced.png\n")

# --- IMAGE 3: Stretch FIRST, THEN Gamma 0.35 ---

if os.path.exists(input_files['img3']):

    img3 = cv2.imread(input_files['img3'], cv2.IMREAD_GRAYSCALE)

    img3 = ensure_uint8(img3)

    # Lift the ultra-dark pixels first, otherwise Gamma(0) = 0
```

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```
    img3_stretched = contrast_stretching(img3, lower_percent=0, upper_percent=100)

    # Now apply Gamma to naturally brighten the visible seeds

    img3_out = gamma_correction(img3_stretched, gamma=0.35)

    cv2.imwrite('image3_enhanced.png', img3_out)

    print(f"✓ Enhanced {input_files['img3']} -> image3_enhanced.png\n")

# --- IMAGE 4: NIGHT CAR - FIXED USING CLAHE LOCAL BRIGHTENING ---

if os.path.exists(input_files['img4']):

    img4 = cv2.imread(input_files['img4'], cv2.IMREAD_GRAYSCALE)

    img4 = ensure_uint8(img4)

    # Step 1: Global Stretch (Clip top 1% to prevent the bright lamp from ruining the scale)

    img4_stretched = contrast_stretching(img4, lower_percent=0, upper_percent=99)

    # Step 2: CLAHE (The FIX). It looks at 8x8 blocks of the car.

    # Since the car is black, CLAHE forces those blocks to stretch into visible grays.

    img4_clahe = clahe_enhancement(img4_stretched, clip_limit=4.0, grid_size=(8, 8))

    # Step 3: Mild Gamma to get a natural aesthetic tone

    img4_out = gamma_correction(img4_clahe, gamma=0.6)

    # Step 4: Unsharp Mask for crisp edges

    blurred = cv2.GaussianBlur(img4_out, (0, 0), 1)

    img4_out = cv2.addWeighted(img4_out, 1.0 + 1.0, blurred, -1.0, 0)

    cv2.imwrite('image4_enhanced.png', img4_out)

    print(f"✓ Enhanced {input_files['img4']} -> image4_enhanced.png (CLAHE local fix)\n")
```

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```
# --- IMAGE 5: CT Scan (CLAHE) ---

if os.path.exists(input_files['img5']):

    img5 = cv2.imread(input_files['img5'], cv2.IMREAD_GRAYSCALE)

    img5 = ensure_uint8(img5)

    # Lift slightly, then CLAHE to reveal organs clearly

    img5_stretched = contrast_stretching(img5, lower_percent=1, upper_percent=99)

    img5_out = clahe_enhancement(img5_stretched, clip_limit=2.0, grid_size=(8, 8))

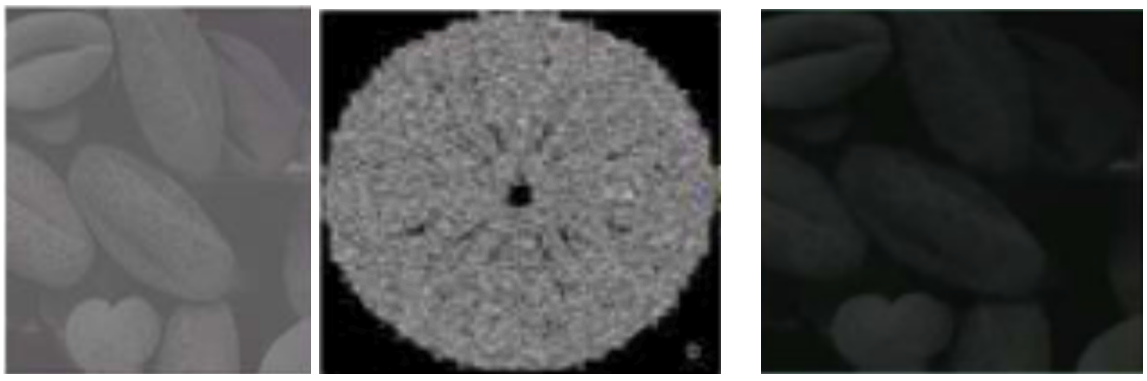
    cv2.imwrite('image5_enhanced.png', img5_out)

    print(f"✓ Enhanced {input_files['img5']} -> image5_enhanced.png\n")

print("--- All images processed successfully! Check your folder. ---")
```

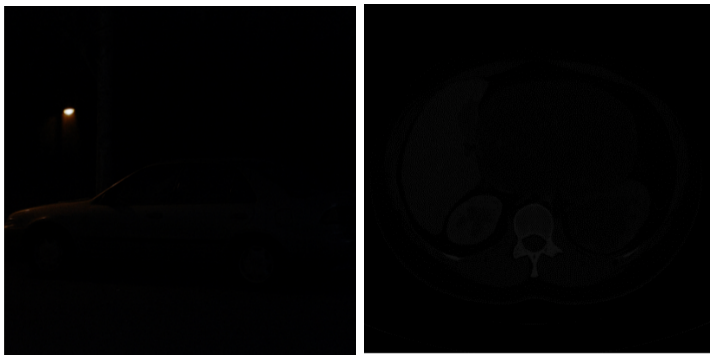
Input images:-

Image 1 Image 2 Image 3



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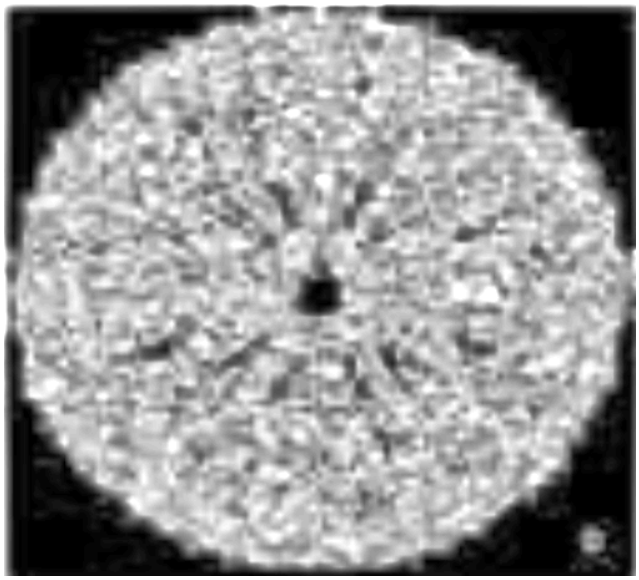
Image 4 Image 5



Output:-

Image 1 Image 2





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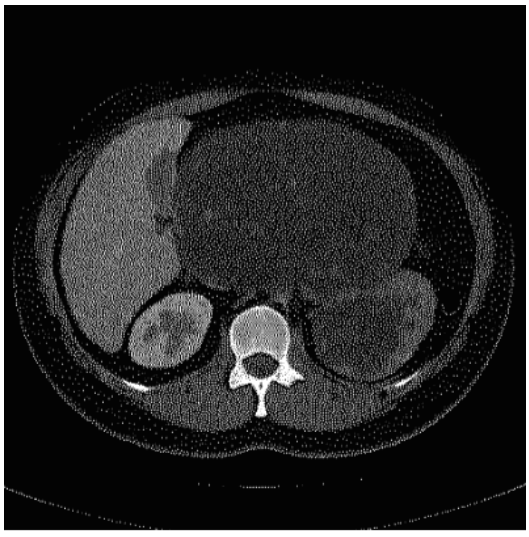
Image 3



Image 4



Image 5



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Comparative Analysis – Effectiveness, Strengths, and Weaknesses

Summary of Effectiveness

Each image enhancement technique works best for a particular type of image problem:

- **Contrast Stretching:** Worked well for the washed-out seed image (Image 1). It expanded the limited range of gray values across the full 0–255 range, making the image clearer and improving its contrast.
- **Gamma Correction:** Was effective for the dark seed image (Image 3) and the nighttime car image (Image 4). It improved the visibility of dark areas by changing brightness in a non-linear way, which is similar to how the human eye responds to different brightness levels.
- **Histogram Equalization:** Worked well for the flat circular pattern (Image 2) and the dark CT scan (Image 5). It helped reveal textures and anatomical details that were difficult to see in the original images.

Important Limitations

Although these techniques can improve an image, they also have some drawbacks.

1. Global Processing Limitation

All three techniques mainly treat the image as a whole. They do not fully understand the difference between individual regions of an image.

For example, in the nighttime car scene, making the dark car visible can cause the bright streetlamp to lose some of its internal details.

This is a natural trade-off: improving one part of the image may reduce the quality of another part.

2. Increase in Noise

Image enhancement can also make existing noise more noticeable.

The reason is that when the intensity values are increased or stretched, the background sensor noise can also be amplified along with the actual image information.

As a result, images such as Image 2 and Image 3 may appear more grainy after enhancement.

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3. Artifacts in Medical Images

Histogram Equalization can sometimes create unwanted patterns or artifacts, especially in medical scans.

For example, in Image 5, the originally black background of the CT scan may become gray and contain visible noise after standard histogram equalization. This can be undesirable because medical images need to preserve meaningful anatomical information without introducing misleading patterns.

Conclusion

The main purpose of this work was to apply gray-level point-processing techniques to five different degraded images and determine which enhancement method works best for each one.

The process involved:

1. Examining the image and its histogram to understand the type of degradation.
2. Identifying the main problem, such as low contrast, severe darkness, or a wide dynamic range.
3. Selecting the appropriate mathematical technique for that particular problem.
4. Comparing the enhanced image with the original to evaluate the improvement.
5. Studying the limitations of each technique, including noise amplification and unwanted artifacts.

Technique Selected for Each Image

Image 1 – Washed-Out Seeds

- **Problem:** Low contrast with gray values concentrated in a small range.
- **Technique:** Linear Contrast Stretching.
- **Reason:** It spreads the limited gray-level range across 0–255, improving the overall contrast.

Image 2 – Circular Noisy Pattern

- **Problem:** Very compressed histogram with hidden textures.
- **Technique:** Histogram Equalization (HE).
- **Reason:** The CDF is used to redistribute the gray levels over a wider range, making hidden patterns and textures more visible.

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Image 3 – Dark Seeds

- **Problem:** Most pixels are concentrated near 0, making the image severely underexposed.
- **Technique:** Power-Law/Gamma Correction with γ**=0.35**.
- **Reason:** A gamma value below 1 brightens the dark regions and helps reveal details hidden in the shadows.

Image 4 – Nighttime Car Scene

- **Problem:** Extremely high dynamic range, with a nearly black car and a very bright streetlamp.
- **Technique:** CLAHE + mild Gamma Correction.
- **Reason:** CLAHE improves contrast in local areas and helps reveal details in the dark regions without relying only on a global transformation.

Image 5 – Dark CT Scan

- **Problem:** Incorrect window/level settings caused important anatomical information to appear too dark.
- **Technique:** Histogram Equalization.
- **Reason:** HE spreads the dark gray values and helps reveal structures such as bones and soft tissues.

Overall Conclusion:-

The results show that there is no single enhancement technique that works perfectly for every image. The best method depends on the type of degradation shown by the image and its histogram.

The techniques successfully improved the five images, but they also demonstrated important limitations such as noise amplification, loss of detail, and unwanted artifacts. Therefore, understanding both the mathematics and the visual effect of each technique is important when choosing an image enhancement method.